

Credit Risk Loan Pricing Engine

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Background:	Master's in Finance, Actuarial Science & Data Science
Dataset:	Give Me Some Credit (Kaggle) — 150,000 borrower records
Methods:	Logistic Regression · XGBoost · Isotonic Calibration · Basel PDxLGDxEAD Framework

Executive Summary

This report presents a credit risk pricing model for consumer loans, built using the Basel-aligned Expected Loss framework ($PD \times LGD \times EAD$) — the regulatory standard used by banks and fintech lenders worldwide. The model predicts each borrower's Probability of Default, combines it with documented Loss Given Default and Exposure at Default assumptions, and converts the resulting Expected Loss into a risk-based interest rate. A critical probability calibration issue was identified and corrected during development, and the analysis was benchmarked against a modern machine learning model (XGBoost) throughout.

Metric	Result
Borrowers analyzed	149,999
Default rate	6.68% of borrowers defaulted within two years
Best PD model	XGBoost (ROC-AUC 0.868 vs Logistic Regression 0.858)
Critical fix	Corrected probability overestimation (4.9x) via isotonic calibration
Strongest risk driver	Payment history (past-due incidents), confirmed by both models
Expected Loss	3.23% of total portfolio exposure — in line with industry norms
Pricing range	7.18% (safest decile) to 28.37% (riskiest decile) interest rate

1. Business Problem & Methodology

Lenders must price loans accurately: charging a high enough interest rate to cover expected credit losses, while remaining competitive enough to attract creditworthy borrowers. This project answers the core credit risk question: given a borrower's financial profile, what interest rate should they be charged?

The pricing logic follows the Basel II/III risk framework in six stages:

- 1. Probability of Default (PD)** — predicts the likelihood a borrower defaults within two years (Logistic Regression).
- 2. ML Benchmark** — XGBoost comparison to test whether a more complex model materially improves accuracy.
- 3. Probability Calibration** — ensuring predicted probabilities reflect true default rates, essential for monetary calculations.
- 4. Loss Given Default (LGD)** — the proportion of exposure lost if default occurs, via documented assumptions.
- 5. Exposure at Default (EAD)** — the amount at risk at the point of default.
- 6. Risk-Based Pricing** — Expected Loss ($PD \times LGD \times EAD$) converted into a fair interest rate.

1.1 Dataset

The analysis uses **Give Me Some Credit**, a well-known Kaggle credit scoring dataset containing 150,000 borrower records with financial and credit history features, and a binary indicator of serious delinquency within two years. The dataset required substantial cleaning: missing income and dependents data, impossible values (age = 0), extreme outliers in utilization and debt ratio, and a documented data-entry anomaly (error codes 96/98) in the payment history fields.

2. Probability of Default Model Results

A clear, monotonic relationship was found between payment history and default risk: borrowers with no past-due incidents default at just 2.7%, rising sharply to 61.2% for borrowers with six or more past-due incidents — a 22x difference.

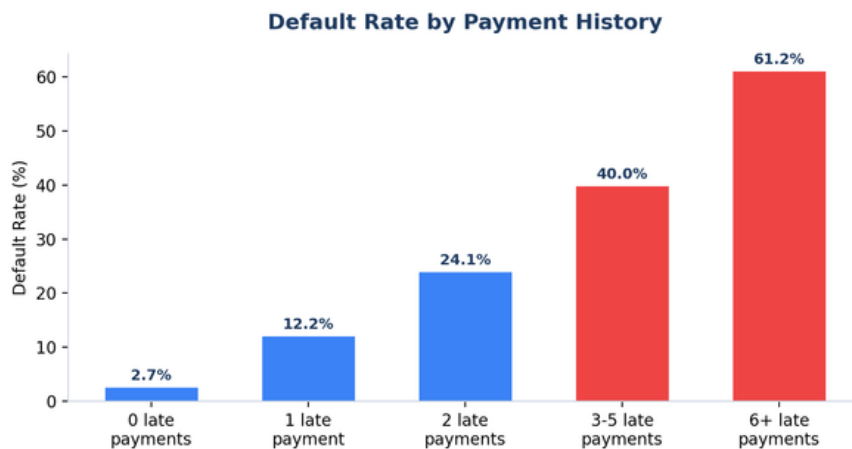


Figure 1 — Observed default rate by payment history.

Logistic Regression achieved a strong ROC-AUC of 0.858, while XGBoost improved this only marginally to 0.868 (~1.2% gain) — the same pattern of modest ML advantage observed consistently across regulated risk pricing tasks. Feature importance from both models independently confirms **payment history** and **credit utilization** as the dominant risk drivers.

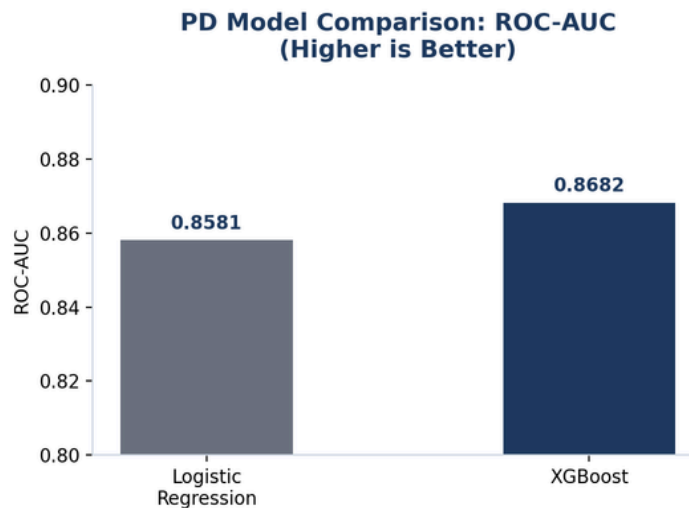


Figure 2 — Model comparison based on ROC-AUC.

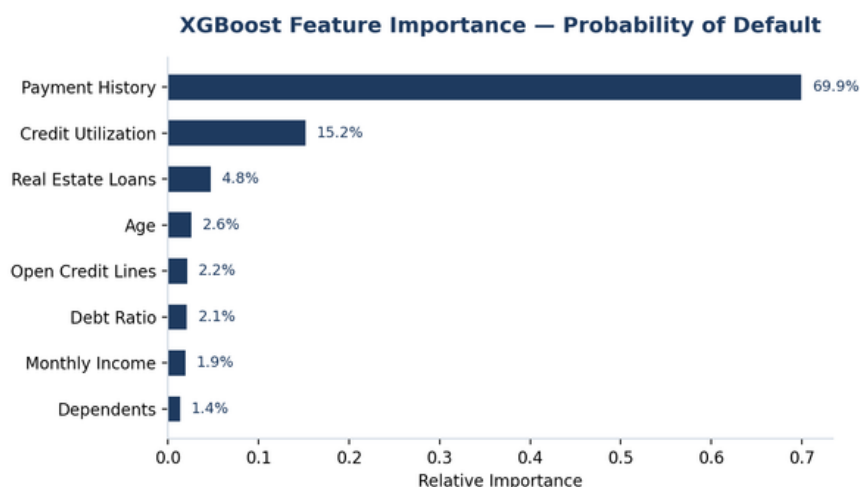


Figure 3 — XGBoost feature importance for default prediction.

3. Critical Finding: Probability Calibration

A significant technical issue was identified during model validation. The initial XGBoost model, trained with class-balancing to address the 6.68% default rate imbalance, achieved excellent discrimination but **severely overestimated actual default probabilities** — predicting a mean PD of 32.3% against an actual rate of 6.68%, a 4.9x inflation. This is a well-known side effect of class-balancing techniques: they improve ranking but distort probability calibration.

Since loan pricing requires an **accurate probability**, not just correct risk ranking, the model was retrained without class-balancing and **isotonic calibration** was applied. This corrected the mean predicted PD to 6.70% (vs actual 6.68%) while slightly improving discrimination (ROC-AUC 0.8685). At the portfolio level, this reduced the Expected-to-Actual Loss ratio from 4.897 to 1.002 — a near-perfect calibration.

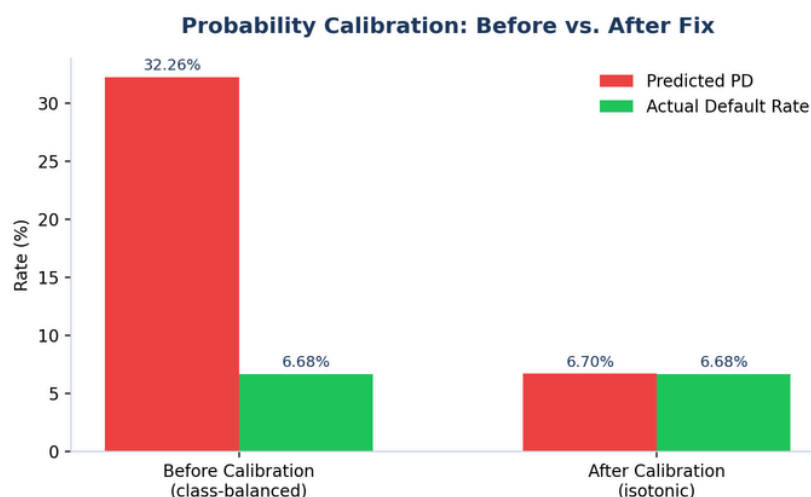


Figure 4 — Predicted vs. actual default rate, before and after calibration.

Why this matters: this calibration step is essential whenever model outputs feed into a monetary calculation (Expected Loss, pricing, capital reserving) rather than a binary accept/reject decision. A model can have excellent discriminative power (high ROC-AUC) while still producing financially unusable probability estimates — a distinction regulated lenders must validate explicitly.

4. Loss Given Default & Exposure at Default

This dataset was designed purely for default prediction and does not include loan amounts, collateral values, or recovery data. Following the same principle of transparent, documented assumptions used for reinsurance loading in comparable insurance pricing work, LGD and EAD were estimated using industry-standard proxies:

- **LGD:** 40% for borrowers with at least one real-estate-backed credit line (partial collateral cushion), 75% for purely unsecured borrowers — consistent with published Basel retail credit benchmarks. Portfolio-weighted average: 53.1%.
- **EAD:** sized as 10x monthly income (a standard affordability-based underwriting convention), capped at €100,000. 13.2% of borrowers reach this cap, reflecting genuinely high-income segments of the portfolio.

Disclaimer: these values are illustrative, documented assumptions — not measured directly from the dataset. Production deployment requires a lender's own historical recovery data to calibrate LGD and actual disbursed amounts for EAD.

5. Risk-Based Pricing Results

Expected Loss was converted into a market-ready interest rate using illustrative industry-standard components: 3% cost of funds, 2% operating cost margin, and 2% profit margin, plus a risk premium equal to the borrower-specific Expected Loss Rate.

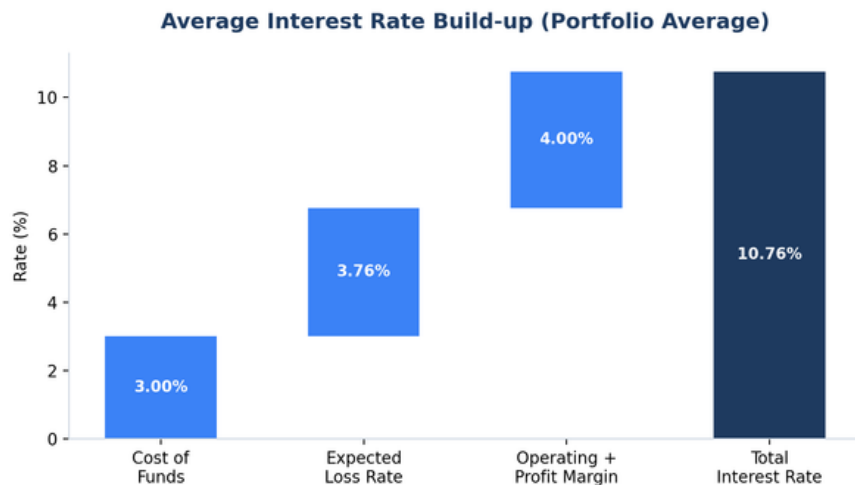


Figure 5 — Average interest rate build-up (portfolio average).

The resulting pricing engine produces a smooth, economically sensible risk gradient: interest rates rise from **7.18%** for the safest risk decile to **28.37%** for the riskiest — a 4x spread directly reflecting each segment's underlying default risk.

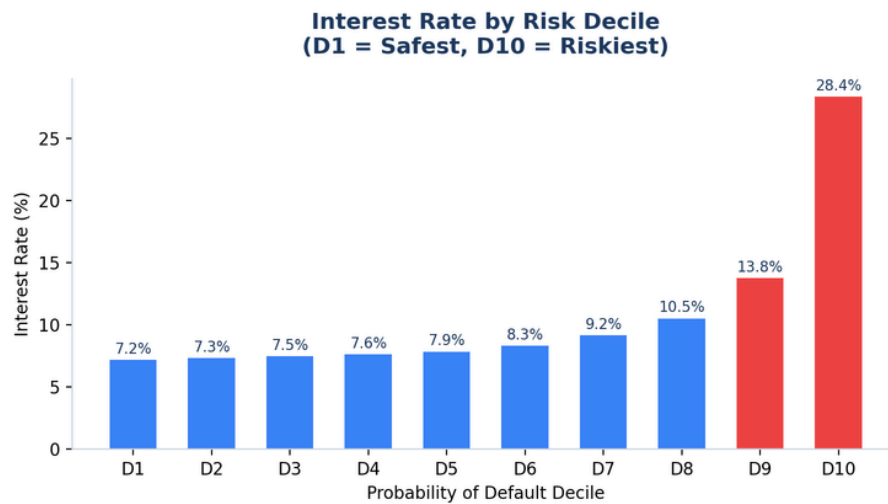


Figure 6 — Interest rate by probability-of-default decile.

Portfolio validation: post-calibration, total Expected Loss (€273.5M) closely matches the realized-default-based estimate (€272.9M) — a ratio of 1.002 — confirming the pricing engine is built on a well-calibrated risk foundation. Expected Loss represents 3.23% of total portfolio exposure, consistent with typical consumer lending loss rates.

6. Conclusion

This project demonstrates a complete, Basel-aligned credit risk pricing pipeline — from raw application data to a risk-differentiated interest rate — combining a rigorously validated statistical model with modern machine learning benchmarking. The explicit identification and correction of a probability calibration issue reflects the kind of validation rigor expected in regulated lending decisions, where predicted probabilities directly drive pricing and capital reserving. The accompanying interactive dashboard allows real-time loan pricing and risk factor exploration for any borrower profile.